AGE 316: MICROCOMPUTER SYSTEMS

LECTURE 15: PROJECT DESIGN & THE FUTURE

Dr. Kadeghe Fue

Department of Agricultural Engineering

Our Journey: From Zero to Hero

1. Basics (Ohm's Law)
2. Hardware (Arduino Uno)
3. Logic (C++ Code)
4. Sensing (Sensors)
5. Acting (Motors)
6. **System Design**

Today: We put it all together into a **Professional Project**.

The Engineering Workflow

Building a successful embedded system follows a strict cycle.



Phase 1: Defining the Problem

Ask the Right Questions

? **What** are we measuring? (Temp, Soil Moisture?)

? **Where** will it live? (Indoor lab or Rainy field?)

? **How** is it powered? (Wall outlet or Battery?)

Example: Smart Greenhouse

Needs to monitor Humidity and Temperature continuously. Needs to control a fan automatically. Must run 24/7 on solar power.

Phase 2: Picking the Right Tools

Controller

Arduino Uno? Too big? Use **Nano**. Needs Wi-Fi? Use **ESP32**.

Sensors

DHT11 is cheap but inaccurate. **SHT31** is pro-grade.

Power

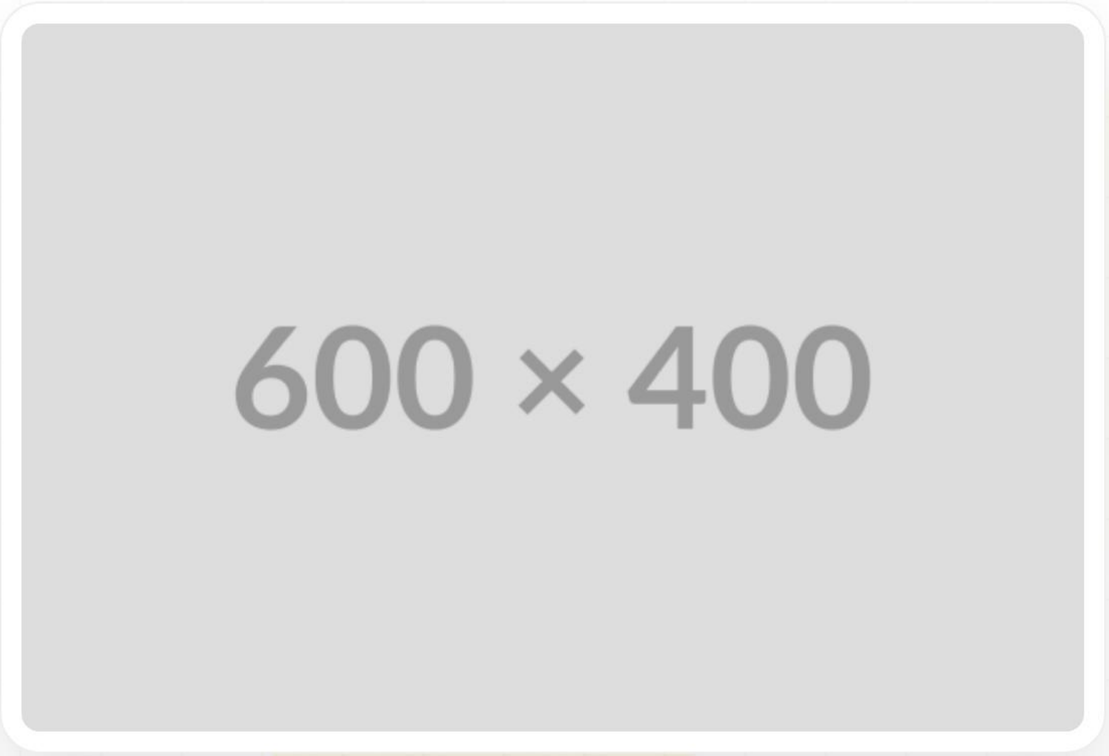
9V Batteries die fast. Use **Li-Ion 18650** or Lead Acid.

Phase 3: Design & Simulation

Don't Solder Yet!

Before building, draw your schematic.

- **Fritzing:** Great for beginners/visuals.
- **EasyEDA / KiCad:** For professional PCB design.
- **Tinkercad:** Simulate code and circuits online.



The #1 Failure: Power



"It works on USB but fails on Battery."

USB (5V)

Great for testing. Limited current (500mA).

Vin Pin (7-12V)

Uses onboard regulator. Gets hot if voltage is too high.

5V Pin (Regulated)

Direct power. Bypasses protection. **Danger zone!**

Doing the Math: Power Budget

Calculate total current draw to choose a battery.

- `Arduino Uno: ~50 mA`
- `LCD Screen: ~20 mA`
- `Relay Module: ~70 mA`
- `Sensors: ~10 mA`
- **`TOTAL: ~150 mA`**

Battery Calculation

Capacity: 2000 mAh (Milliamp-Hours)

$$ \frac{2000 \text{ mAh}}{150 \text{ mA}} \approx 13.3 \text{ Hours} $$

Result: It will run for half a day.

Handling Voltage

Batteries change voltage as they drain. Microcontrollers need steady voltage.

Linear Regulator (7805)

Burns excess voltage as **Heat**.

Inefficient but cheap.

Buck Converter (LM2596)

Converts voltage efficiently using magnetics.

90% Efficiency. Use this for battery projects!



Beyond the Uno: Form Factors

Arduino Nano

Same brain (ATmega328P) but tiny. Breadboard friendly. Perfect for final builds.

Arduino Mega

More Pins (54!), More Memory. Used for complex robots or 3D printers.

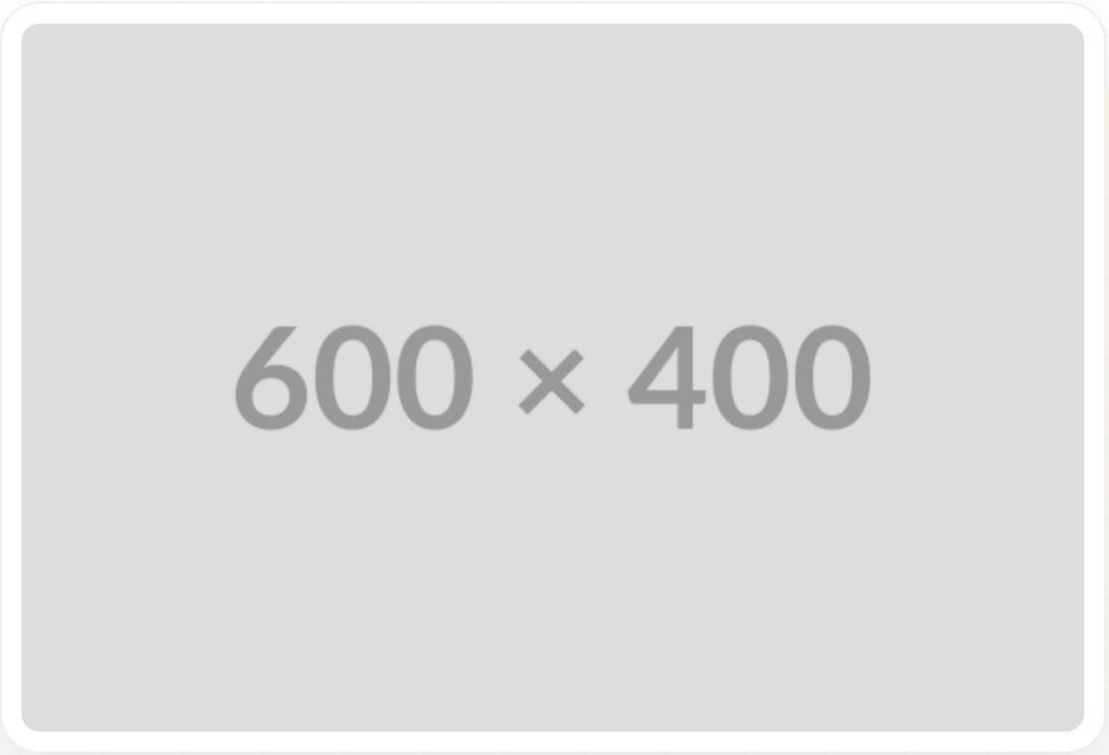
ESP32

The modern king. Faster, Wi-Fi, Bluetooth built-in. 3.3V Logic.

Making it Permanent

Breadboards are for testing. Wires fall out easily.

1. **Perfboard:** Solder components manually on a grid board.
2. **Custom PCB:** Design a board (KiCad), send file to China (JLCPCB), receive professional board.



Protection: Enclosures

In Agriculture, the environment is the enemy (Dust, Rain, Mud).

Project Box

Generic plastic waterproof (IP65) boxes.

3D Printing

Custom fit cases designed in CAD (Fusion 360).

Conformal Coating

Spray-on plastic for the circuit board to resist moisture.

Course Summary

We have covered:

- **Microcontrollers:** Architecture & Logic.
- **Electronics:** V=IR, Dividers, PWM.
- **Programming:** C++, Functions, Libraries.
- **Interfacing:** Sensors, Motors, Displays.
- **Comms:** UART, I2C, SPI, Wi-Fi.

100%

Ready for Ag-Tech

The Future is YOU

"Agriculture 4.0 is not about bigger tractors. It's about smarter data."

Drones

Crop monitoring & Spraying.

Agri-Bots

Autonomous weeding & harvesting.

IoT

Smart irrigation & Soil analysis.

Dr. Fue's Final Advice

1. Start small. Get one LED blinking first.

2. Datasheets are your best friends. Read them.

3. You will burn components. It's part of learning.

4. Comment your code. "Future You" will thank you.

Thank You!



AGE 316

Microcomputer Systems

Sokoine University of Agriculture

Instructor: Dr. Kadeghe Fue

Questions?

"The only stupid question is the one you don't ask."

Good Luck with your Projects!

Image Sources



https://upload.wikimedia.org/wikipedia/en/c/c1/SUA.png

Source: wikipedia.org